

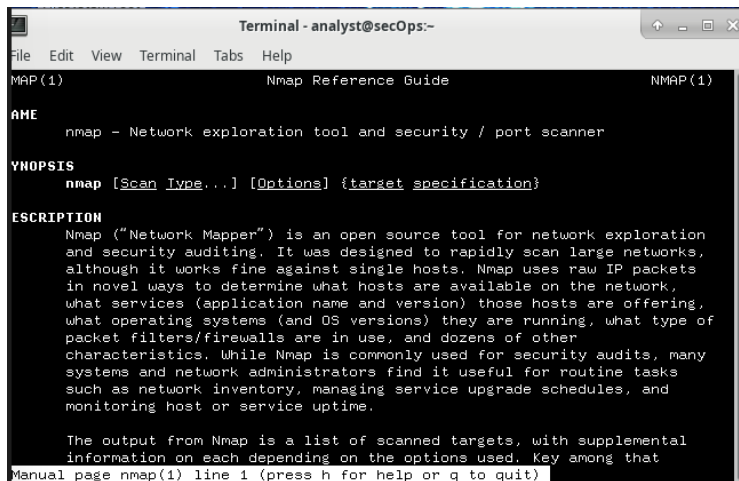
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Materi : Using Nmap

Part 1 Exploring Nmap

Step 1

C. At the terminal prompt, enter `man nmap`.



```
Terminal - analyst@secOps:-
File Edit View Terminal Tabs Help
MAP(1) Nmap Reference Guide NMAP(1)

NAME
    nmap - Network exploration tool and security / port scanner

SYNOPSIS
    nmap [Scan Type...] [Options] {target specification}

DESCRIPTION
    Nmap ("Network Mapper") is an open source tool for network exploration
    and security auditing. It was designed to rapidly scan large networks,
    although it works fine against single hosts. Nmap uses raw IP packets
    in novel ways to determine what hosts are available on the network,
    what services (application name and version) those hosts are offering,
    what operating systems (and OS versions) they are running, what type of
    packet filters/firewalls are in use, and dozens of other
    characteristics. While Nmap is commonly used for security audits, many
    systems and network administrators find it useful for routine tasks
    such as network inventory, managing service upgrade schedules, and
    monitoring host or service uptime.

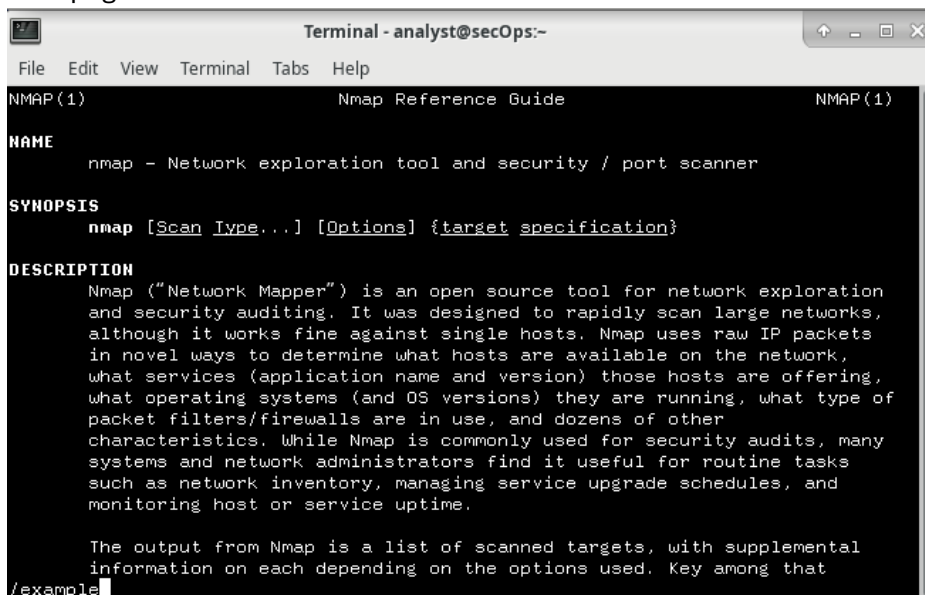
    The output from Nmap is a list of scanned targets, with supplemental
    information on each depending on the options used. Key among that

Manual page nmap(1) line 1 (press h for help or q to quit)
```

What is Nmap? Nmap adalah sebuah tool open source yang digunakan secara luas untuk eksplorasi jaringan, pemindaian keamanan, dan audit sistem

What is nmap used for? Nmap digunakan untuk Mendeteksi berbagai jaringan seperti pemindaian port, menemukan host yang aktif, deteksi sistem operasi, audit keamanan dan kerentanan

d. Type `/example` and press ENTER. This will search for the word example forward through the man page



```
Terminal - analyst@secOps:-
File Edit View Terminal Tabs Help
NMAP(1) Nmap Reference Guide NMAP(1)

NAME
    nmap - Network exploration tool and security / port scanner

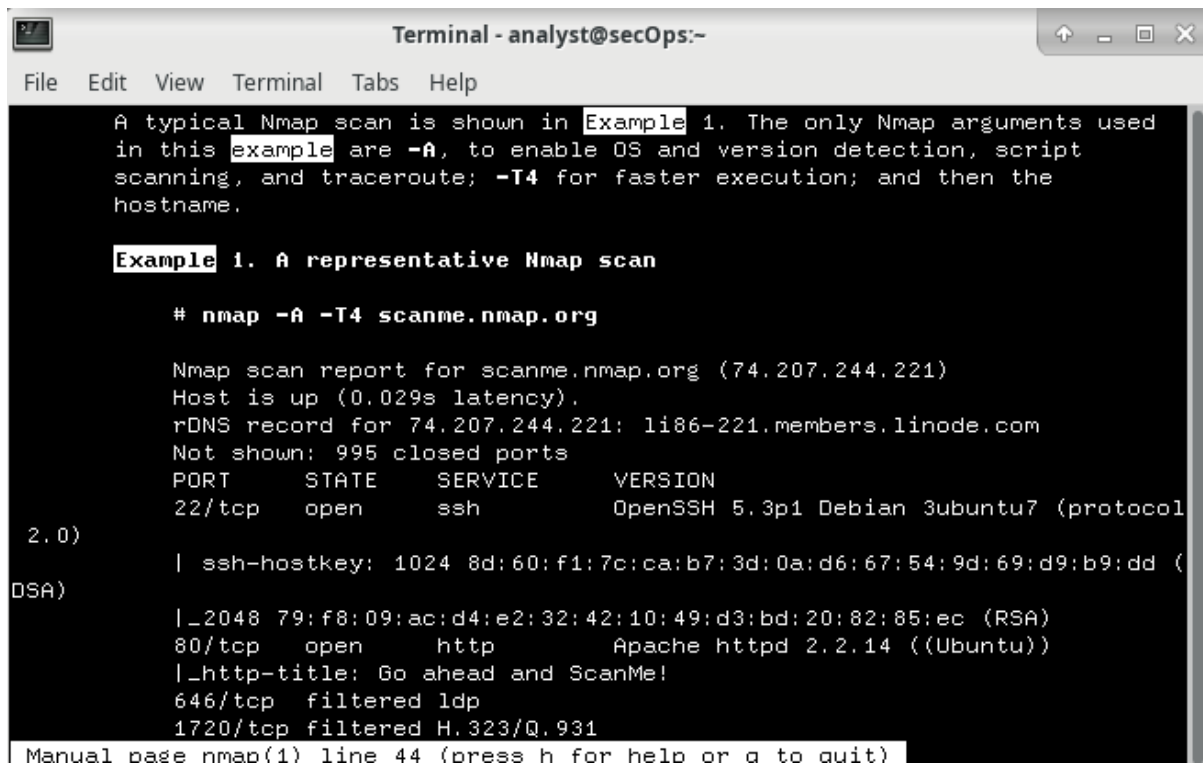
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/example
```

e. In the first instance of example, you see three matches. To move to the next match, press n.



```
Terminal - analyst@secOps:~
File Edit View Terminal Tabs Help

A typical Nmap scan is shown in Example 1. The only Nmap arguments used
in this example are -A, to enable OS and version detection, script
scanning, and traceroute; -T4 for faster execution; and then the
hostname.

Example 1. A representative Nmap scan

# nmap -A -T4 scanme.nmap.org

Nmap scan report for scanme.nmap.org (74.207.244.221)
Host is up (0.029s latency).
rDNS record for 74.207.244.221: li86-221.members.linode.com
Not shown: 995 closed ports
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 5.3p1 Debian 3ubuntu7 (protocol
2.0)
| ssh-hostkey: 1024 8d:60:f1:7c:ca:b7:3d:0a:d6:67:54:9d:69:d9:b9:dd (
DSA)
|_2048 79:f8:09:ac:d4:e2:32:42:10:49:d3:bd:20:82:85:ec (RSA)
80/tcp    open  http         Apache httpd 2.2.14 ((Ubuntu))
|_http-title: Go ahead and ScanMe!
646/tcp   filtered ldp
1720/tcp  filtered H.323/Q.931

Manual page nmap(1) line 44 (press h for help or q to quit)
```

Look at Example 1. What is the nmap command used? Perintah nmap yang digunakan -A digunakan untuk mengaktifkan beberapa fitur deteksi dan -T4 digunakan untuk mempercepat proses eksekusi pemindaian.

Use the search function to answer the following questions. What does the switch -A do? -A digunakan untuk mengaktifkan aggressive scan

What does the switch -T4 do? -T4 digunakan untuk mengatur kecepatan dan intensitas pemindaian. Secara spesifik, -T4 memilih mode **Aggressive**, yang berfungsi untuk mempercepat proses pemindaian jaringan agar selesai lebih cepat.

Part 2 Scanning your localhost

Step 1 scan your localhost

A **enter nmap -A -T4 localhost**

```
Terminal - analyst@secOps:~
File Edit View Terminal Tabs Help

[analyst@secOps ~]$ man nmap
[analyst@secOps ~]$ nmap -A -T4 localhost
Starting Nmap 7.70 ( https://nmap.org ) at 2026-05-31 08:06 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000035s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_--rw-r--r--  1 0      0      0 Mar 26  2018 ftp_test
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 127.0.0.1
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     At session startup, client count was 1
|     vsFTPD 3.0.3 - secure, fast, stable
|_End of status
|   STAT:
|   FTP server status:
|     Connected to 127.0.0.1
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     At session startup, client count was 1
|     vsFTPD 3.0.3 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 7.7 (protocol 2.0)
| ssh-hostkey:
|   2048 b4:91:f9:f9:d6:79:25:86:44:c7:9e:f8:e0:e7:5b:bb (RSA)
|   256  06:12:75:fe:b3:89:29:4f:8d:f3:9e:9a:d7:c6:03:52 (ECDSA)
|_  256  34:5d:f2:d3:5b:9f:b4:b6:08:96:a7:30:52:8c:96:06 (ED25519)
Service Info: Host: Welcome

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.17 seconds
[analyst@secOps ~]$ if config
```

Which ports and services are opened? Total port yang dipindai pada target localhost 127.0.0.1 terdapat 1 port yaitu port 21/tcp dengan service layanan ftp dan 22/tcp dengan service layanan SSH

For each of the open ports, record the software that is providing the services? Untuk port terbuka yaitu port 21/tcp yang dijalankan oleh perangkat lunak vsftpd pada kolom version, terdeteksi versi vsftpd 2.0.8 or later dan port 22/tcp yang dijalankan oleh OpenSSH 7.7 (protocol 2.0)

What is the operating system? Baris service info: Host: welcome merupakan indikasi penamaan host banner khas mesin linux

Step 2 Scan your network

```
Terminal - analyst@secOps:~  
File Edit View Terminal Tabs Help  
> ^C  
[analyst@secOps ~]$ ifconfig  
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 fd17:625c:f037:2:a00:27ff:fe17:99c9 prefixlen 64 scopeid 0x0<global>  
bal>  
    inet6 fe80::a00:27ff:fe17:99c9 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:17:99:c9 txqueuelen 1000 (Ethernet)  
    RX packets 5 bytes 1445 (1.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 13 bytes 1587 (1.5 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 2254 bytes 127338 (124.3 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2254 bytes 127338 (124.3 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Record the IP address and subnet mask for your VM. Which network does your VM belong to?
IP address 10.0.2.15 dan subnet 255.255.255.0

b. enter nmap -A -T4 network address/prefix

```
Terminal - analyst@secOps:~  
File Edit View Terminal Tabs Help  
[analyst@secOps ~]$ nmap -A -T4 10.0.2.0/24  
Starting Nmap 7.70 ( https://nmap.org ) at 2026-05-31 08:08 EDT  
Nmap scan report for 10.0.2.15  
Host is up (0.000028s latency).  
Not shown: 998 closed ports  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 2.0.8 or later  
| ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ -rw-r--r--  1 0      0      0 Mar 26  2018 ftp_test  
| ftp-syst:  
|   STAT:  
|   FTP server status:  
|     Connected to 10.0.2.15  
|     Logged in as ftp  
|     TYPE: ASCII  
|     No session bandwidth limit  
|     Session timeout in seconds is 300  
|     Control connection is plain text  
|     Data connections will be plain text  
|     At session startup, client count was 6  
|     vsFTPD 3.0.3 - secure, fast, stable  
|_ End of status  
22/tcp    open  ssh      OpenSSH 7.7 (protocol 2.0)  
| ssh-hostkey:
```

```

22/tcp open  ssh      OpenSSH 7.7 (protocol 2.0)
| ssh-hostkey:
| 2048 b4:91:f9:d6:79:25:86:44:c7:9e:f8:e0:e7:5b:bb (RSA)
| 256 06:12:75:fe:b3:89:29:4f:8d:f3:9e:9a:d7:c6:03:52 (ECDSA)
|_ 256 34:5d:f2:d3:5b:9f:b4:b6:08:96:a7:30:52:8c:96:06 (ED25519)
Service Info: Host: Welcome

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (1 host up) scanned in 26.87 seconds
[analyst@sec0ps ~]$

```

How many hosts are up? Jumlah host yang aktif 1 ditandai dengan Nmap done: 256 IP addresses (1 host up) scanned

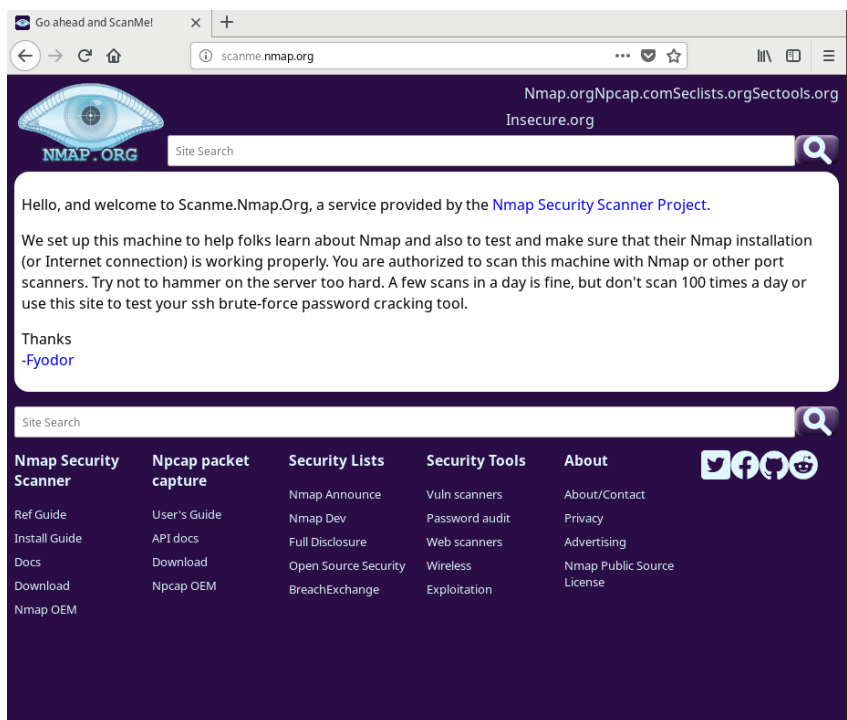
From your Nmap results, list the IP addresses of the hosts that are on the same LAN as your VM. List some of the services that are available on the detected hosts.

Meskipun Nmap memindai seluruh blok sub-jaringan (256 IP addresses), hanya ada satu perangkat aktif yang terdeteksi berada di dalam LAN yang sama dengan VM Anda:

- **IP Address Target: 10.0.2.15** (Ini merupakan IP dari host/VM itu sendiri atau perangkat lain di subnet lokal tersebut yang merespons).

Layanan service yang tersedia FTP dan SSH dengan port 21 dan 22

Step 3 Scan a remote server



What is the purpose of this site? Tujuannya Membantu orang-orang belajar tentang penggunaan Nmap. Menyediakan sarana uji coba untuk memastikan bahwa instalasi Nmap (atau koneksi internet pengguna) berfungsi dengan benar. Pengguna diberikan izin resmi untuk memindai mesin tersebut menggunakan Nmap atau pemindai port lainnya secara legal, dengan catatan tidak memindainya secara berlebihan (tidak sampai ratusan kali sehari atau menggunakannya untuk uji coba peretasan kata sandi *brute-force* SSH).

```
[analyst@secOps ~]$ nmap -A -T4 scanme.nmap.org
Starting Nmap 7.70 ( https://nmap.org ) at 2026-06-31 08:10 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered ports
PORT      STATE SERVICE        VERSION
22/tcp    open  ssh            OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http           Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo     Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 177.92 seconds
[analyst@secOps ~]$
```

Which ports and services are opened?

Port 22/tcp dengan Service: ssh (OpenSSH 6.6.1p1 Ubuntu)

Port 80/tcp dengan Service: http (Apache httpd 2.4.7)

Port 9929/tcp dengan Service: nping-echo (Nping echo)

Port 31337/tcp dengan Service: tcpwrapped

Which ports and services are filtered?

Terdapat **996 port** yang berstatus difilter (**996 filtered ports**). Port-port ini tidak merespons karena dilindungi oleh kebijakan firewall atau perangkat keamanan pada server target.

What is the IP address of the server?

IP Address (IPv4): 45.33.32.156 (Tertera pada baris Nmap scan report for scanme.nmap.org (45.33.32.156)).

IP Address (IPv6): 2600:3c01::f03c:91ff:fe18:bb2f (Tertera pada baris Other addresses for scanme.nmap.org).

What is the operating system

Sistem operasi server tersebut adalah **Linux** (Tertera secara spesifik pada baris Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel). Karakteristik distribusinya juga diperkuat oleh kemunculan sistem operasi **Ubuntu Linux** pada detail banner port 22.

REFLECTION

Nmap sangat berkontribusi dalam meningkatkan keamanan jaringan karena bertindak sebagai alat audit proaktif bagi para administrator sistem. Dalam aspek pertahanan, Nmap membantu tim keamanan untuk memetakan seluruh aset yang terhubung ke jaringan, mengidentifikasi port mana saja yang terbuka secara tidak sengaja, dan mendeteksi layanan atau aplikasi usang yang memiliki celah keamanan. Dengan memanfaatkan fitur Nmap Scripting Engine, administrator dapat melakukan simulasi kerentanan secara mandiri untuk menemukan

kelemahan sistem sebelum dieksploitasi oleh pihak luar. Selain itu, alat ini juga berguna untuk memverifikasi apakah aturan pada *firewall* atau sistem pertahanan jaringan sudah bekerja dengan optimal dalam memblokir akses yang tidak sah.

Sebaliknya, karakteristik Nmap yang mampu mengungkap struktur mendalam suatu jaringan juga menjadikannya alat yang sangat berbahaya di tangan aktor ancaman. Sebelum melancarkan serangan utama, peretas menggunakan Nmap untuk tahap pengintaian guna mengumpulkan intelijen mengenai target. Dengan informasi mengenai port yang terbuka, versi aplikasi yang berjalan, serta jenis sistem operasi yang digunakan, aktor ancaman dapat memetakan kelemahan target secara presisi. Informasi tersebut kemudian digunakan untuk memilih jenis *exploit* atau malware yang paling efektif untuk menyusup ke dalam sistem, melakukan serangan *brute-force* pada port layanan seperti SSH, atau bahkan melumpuhkan jaringan dengan membanjiri target menggunakan paket pemindaian yang agresif.